

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

## University Examination

Date: 7/03/2018 Day: Wednesday

Time: 10.00 am to 10.30 am

Total Marks: 20

|                           |  |  |  |  |  |  |  |  |  |
|---------------------------|--|--|--|--|--|--|--|--|--|
| Student's ID No.          |  |  |  |  |  |  |  |  |  |
|                           |  |  |  |  |  |  |  |  |  |
| Student's Name: _____     |  |  |  |  |  |  |  |  |  |
| Student's Sign : _____    |  |  |  |  |  |  |  |  |  |
| Supervisor's Name : _____ |  |  |  |  |  |  |  |  |  |
| Supervisor's Sign: _____  |  |  |  |  |  |  |  |  |  |

|                                                                                          |
|------------------------------------------------------------------------------------------|
| Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following |
| Examination : BPT/MPT- March 2018                                                        |
| Semester: [1]                                                                            |
| Section : I                                                                              |
| Course Code : BC 102                                                                     |
| Course Name: BIOCHEMISTRY                                                                |

### Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes. Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

### SECTION-I

**Multiple Choice Questions: (MCQs)**  
(Encircle the correct answer)

(10 × 1 = 10)

**1. Carbohydrates are**

- a. Poly Hydroxy Amino acids.
- b. Poly Hydroxy esters of Fatty acids.
- c. Poly Hydroxy Aldose Ketose Saccharides.
- d. None of the above.

**2. The third carbon in any fatty acid chain structure is referred as \_\_\_\_\_**

- a. Alpha Carbon
- b. Beta Carbon
- c. Omega Carbon
- d. Gamma Carbon

**3. Gout is a disorder which \_\_\_\_\_ .**

- a. Is caused due to some issues in purine metabolism
- b. Is caused due to accumulation of uric acid in joints
- c. Can be treated by NSAIDs
- d. All of the above

4. 37year old obese male fainted due to fatigue, Blood biochemistry revealed Ketoacidosis. The patient is administered with Omega3 FA and was found to get ameliorated with health. Which biochemical cycle defect explains such condition?
  - a. Aerobic Glycolysis
  - b. Anaerobic Glycolysis
  - c. Beta Oxidation
  - d. Omega Oxidation
5. 3 month infant was observed to have failure in glucose uptake in the body, due to some congenital defect. Which enzyme is basically found to be defected in such condition?
  - a. Hexokinase
  - b. Glucokinase
  - c. Phosphoglucoisomerase
  - d. None of the above
6. The rare type of thalassemia is \_\_\_\_\_ thalassemia.
  - a. Alpha
  - b. Beta
  - c. Gamma
  - d. Delta
7. Maltose is \_\_\_\_\_
  - a. Glucose Glucose Disaccharide
  - b. Glucose Galactose Disaccharide
  - c. Glucose Sucrose Disaccharide
  - d. Glucose Fructose Disaccharide
8. The Enzyme Catalyzes the reaction from Alpha KetoGlutarate to Succinyl CoA is?
  - a. Alpha Keto Glutarate Hydratase
  - b. Alpha Keto Glutarate Oxidase
  - c. Alpha Keto Glutarate Dehydrogenase
  - d. Hexokinase
9. Which enzyme is responsible for the transfer of fatty acid intermediate to mitochondria for beta oxidation
  - a. Acyl Carnitine Transferase
  - b. Acetyl Carnitine Transferase
  - c. Acyl Carnitine Transenolase
  - d. None of the above
10. The range of Fasting Blood Sugar (FBS/FPG) is \_\_\_\_\_
  - a. 70-110 mg/dl
  - b. 69-109 mg/dl
  - c. 80-140 mg/dl
  - d. > 180 mg/dl

**Identify the sentence as True or False:**

**(Please tick mark the answer)**

11. Arginine and Histidine are hydrophobic amino acid
12. Inosine Mono Phosphate is intermediate for deNovo Pathway
13. Hemophilia occurs in both sexes
14. Insulin is a Hyper glycemic Hormone
15. Diabetes is a chronic condition occurs in hypoglycemia

**(5 × 1 = 5)**

**True False**

|                          |                          |
|--------------------------|--------------------------|
| <input type="checkbox"/> | <input type="checkbox"/> |
| <input type="checkbox"/> | <input type="checkbox"/> |
| <input type="checkbox"/> | <input type="checkbox"/> |
| <input type="checkbox"/> | <input type="checkbox"/> |
| <input type="checkbox"/> | <input type="checkbox"/> |

Match the following

(5 × 1 = 5)

|                                                                                                             |  |                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------|
| 16. Kinase<br>17. Purine<br>18. Lactic acid to Pyruvate<br>19. Mucopolysaccharadosis<br>20. Omega Oxidation |  | A. Triacyl Glyceraldehyde<br>B. Adenine<br>C. Glucose<br>D. Uric Acid<br>E. 7 Types<br>F. Cori Cycle<br>G. Transferase |
|-------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------|

(16 -    ),    (17-    ),    (18-    ),    (19-    ),    (20-    )

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## I Semester of B.P.T. Examination

March 2018

### BC 102 BIOCHEMISTRY

Date: 7/03/2018

Time: 10.30 am to 1.00 pm

Maximum Marks: 50

#### Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

#### SECTION-II

**Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)**

- a. Define Carbohydrates? State Its Classification?
- b. Define Balanced Diet and RDA for daily meal?
- c. What is Omega Oxidation & Peroxisomal Oxidation?
- d. Explain Haemoglobinopathies?
- e. Define the term “Glucose Tolerance.”
- f. Explain Protein Energy Malnutrition with examples.
- g. Explain Classification of Enzymes.

**Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)**

- a. Draw a schematic diagram of cell and explain the functions of organelles in brief.
- b. Explain the mode of Enzyme Catalysis.
- c. Explain the mode of Inhibition of Enzymes.
- d. Explain the entire carbohydrate Metabolism Diagrammatically
- e. Explain Blood Glucose Homeostasis

**Q. 3. Essay (2 out of 3) (2 × 10 = 20)**

- a. Justify why Tricarboxylic Acid Cycle is known as Central Metabolic Pathway and Explain.
- b. Explain Beta Oxidation and Carnitine Shuttle
- c. Explain Erythroblastosis Foetalis?